

Referee package (DRAFT) – the enumerated Reading-H selection margin upgraded to a controlled-error, node-free no-condensate statement

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This self-contained referee document concerns the ESTIMATOR-UPGRADE sub-proof of B1-RH-ENUM: the upgrade of the estimator-grade Reading-H selection $\Delta F > 0$ to a CONTROLLED-ERROR, node-free (continuum) no-condensate statement, across all numerical quadrature knobs and for a multi-shell ensemble’s exact (off-diagonal-inclusive) free energy. **Scope:** the enumerated competitor ensemble at the operating point; single-shell SMA with the $(M, \delta I, \text{amplitude-grid})$ knobs. The competitor generation uses the single-shell SMA knobs, while the validation free energy is evaluated by the multi-shell exact off-diagonal-inclusive engine. **Not claimed:** this is not the full $\mathcal{C}_{\text{full}}$ selection (that is the Reading-H package); the folder result is T4 STRONG EVIDENCE, not a theorem.

2. The result

For the enumerated Reading-H ensemble, the estimator-grade $\Delta F > 0$ is promoted to a node-free no-condensate certificate by combining (a) grid-independent curvature certification at two quadrature resolutions, (b) a curvature-chord interval lower bound $\Delta F \geq \min(v_i, v_{i+1}) - \frac{1}{8}M_i\delta^2$, and (c) a near-reference positive-definite Hessian covering the thin region where the off-diagonal correction is $O(\text{amplitude}^4)$. The exact (slogdet) engine is re-used by neuter-import of a frozen legacy script, validated against its recorded outputs to 5×10^{-8} .

3. Structure of the argument

(i) The estimator ΔF is reproduced at two quadrature resolutions and the curvature is certified grid-independent. (ii) The interval lower bound makes the no-condensate conclusion robust to discretisation. (iii) The exact-Wick off-diagonal bracket is shown $O(\text{amplitude}^4)$ near the reference, so it never shifts the (0,0) Hessian, which stays equal to the diagonal one. (iv) A physical secondary-shell ratio is shown to lie interior to a region already certified condensate-free. The knob-completeness (single-shell SMA, amplitude grid, trial mass) is exercised script-by-script.

4. Numerical reproduction

The reproduction surface is the following explicit manifest of 22 scripts (each carries self-test asserts and exits 0 iff all pass); the union covers the controlled-error certification, the knob sweeps, and the frozen-engine neuter-import validation. Run each as `python <path>; compare against expected/`.

```

codes/vacuum/estimator_upgrade_enumerated.py
codes/vacuum/estimator_upgrade_knobs.py
codes/vacuum/g3pb3_ratio_extraction.py
codes/vacuum/res_endpoint_admissible_pin.py
codes/vacuum/res5_rpa_screening.py
codes/vacuum/res5_a0_skeleton_sensitivity.py
codes/vacuum/res5_allorder_commonmode.py
codes/vacuum/res5_017_chi_bypass.py
codes/vacuum/res5_commonmode_envelope.py
codes/vacuum/res5_dr2_kappa_bound.py
codes/vacuum/res5_dressed_loop_parameter.py
codes/vacuum/res5_endpoint_2pi_bound.py
codes/vacuum/res5_higherloop_skeleton.py
codes/vacuum/res5_2pi_commonmode_monotonicity.py
codes/vacuum/res5_oneloop_disentangle.py
codes/vacuum/res5_orderedbcc_parallel.py
codes/vacuum/res5_projection_factor.py
codes/vacuum/res5_sunset_norm_map.py
codes/vacuum/res5_susceptibility_ratio.py
codes/vacuum/res5_tail_budget.py
codes/vacuum/twoshell_anchored_bracket.py
codes/vacuum/twoshell_continuum_bound.py

```

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.
- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder’s bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder’s scripts (all asserts pass).

6. Result footer

Result ID:	B1-RH-ENUM / ESTIMATOR-UPGRADE referee package (DRAFT)
Precise statement:	The enumerated Reading-H estimator-grade dF>0 is upgraded to a controlled-error, node-free (continuum) no-condensate statement across all quadrature knobs and for the multi-shell exact free energy, via two-resolution curvature certification + curvature-chord interval bound + near-reference PD Hessian + frozen-engine neuter-import (validated 5e-8).
Scope:	enumerated competitor ensemble, operating point, single-shell SMA knobs. T4 STRONG EVIDENCE.
Evidence grade:	controlled-error continuum no-condensate (T4);

EXECUTED (folder scripts PASS).

Reproduction command: python codes/vacuum/estimator_upgrade_*.py ;
res_endpoint_admissible_pin.py ; ...

Falsifier: an enumerated competitor with $dF \leq 0$ surviving
the controlled-error bound; or a knob setting
that flips the curvature certificate.

Tier before / after: DRAFT referee package; folder result T4. NOT
operator-confirmed.

No-overclaim statement: controlled-error STRONG EVIDENCE within the
enumerated-ensemble scope; not the full C_{full}
selection, not a theorem.

Next required action: operator review + confirm -> then build the
PUBLISHED bundle.